NFSA Monthly Allocation, Transaction and Foodgrain Distribution Dashboard

1. Project Objective:

The objective of this project is to analyse and visualize the monthly allocation, transactions, and foodgrain distribution under the National Food Security Act (NFSA) across Indian states and union territories. The dashboard aims to monitor the efficiency of foodgrain allocation and distribution, evaluate state-wise performance, identify distribution gaps, and provide actionable insights that support transparent and data-driven decision-making for the Public Distribution System (PDS).

2. Data Source:

Dataset Original Source Link: https://ndap.niti.gov.in/dataset/7115

Timeline: 2017 - 2024

3. Problem Statement:

The **National Food Security Act (NFSA)** aims to ensure food security by providing subsidized foodgrains to eligible beneficiaries through the Public Distribution System (PDS). However, monitoring the efficiency of foodgrain allocation, distribution, and beneficiary transactions across different states and over multiple months is challenging due to the large volume of data and the lack of a centralized analytical view.

Without an interactive reporting system, it is difficult to identify disparities between allocated and distributed foodgrains, evaluate the adoption of digital distribution methods such as ePoS and Aadhaar authentication, detect underperforming regions, and monitor overall distribution efficiency. This limits the ability of policymakers and administrators to make timely, data-driven decisions.

4. Attributes:

Attribute Name	Description
Source Year	The original year in which the data was published or collected from the NFSA/PDS source.
State Name	Name of the Indian State or Union Territory where the foodgrain allocation and distribution activities took place.
Region Name	Geographical region of the state (e.g., North, South, East, West, North-East, Central) used for regional analysis.
Source Month	The original month from the source dataset representing the reporting period.
Food grains allocated	Total quantity of foodgrains allocated to the state under the National Food Security Act (NFSA) for the specified month, usually measured in metric tonnes (MT).
Ration cards issued	Total number of ration cards issued to eligible beneficiaries in the state under the Public Distribution System (PDS).
Transaction for ration cards	Total number of ration card transactions recorded during the reporting month.
Aadhaar authenticated Transactions	Number of ration card transactions successfully authenticated using Aadhaar-based biometric verification.

Aadhaar Authenticated Transaction (%)	Percentage of total ration card transactions that were authenticated through Aadhaar. It indicates the adoption of digital authentication.
ePoS (Electronic Point of Sale system) distribution of food grains	Quantity of foodgrains distributed through Electronic Point of Sale (ePoS) devices, enabling digital and transparent beneficiary verification.
Manual distribution of food grains	Quantity of foodgrains distributed manually without using ePoS devices, generally due to connectivity or operational constraints.
Distribution of food grains	Total quantity of foodgrains distributed to beneficiaries during the reporting month, including both ePoS and manual distribution.
YearCode	Numeric code representing the reporting year, used for sorting and time-based analysis (e.g., 2024, 2025).
Year	Reporting year of the transaction and distribution data.
MonthCode	Numeric representation of the month (1 = January, 2 = February, ..., 12 = December) used for chronological sorting in reports.
Month	Name of the reporting month (e.g., January, February, March) used for trend analysis and visualization.

5. Tools & Technologies:

Microsoft Excel

- Data cleaning
- Handling missing values
- Data preprocessing
- Initial data analysis

Power Query

- Data transformation
- Data type formatting
- Removing duplicates
- Creating calculated columns

Power BI

- Data modeling
- DAX (Data Analysis Expressions) calculations
- Dashboard creation
- Interactive visualizations

6. Data Pre-Processing (Excel / Power Query)

- **Excel:** Data Cleaning and Transformation – Changed column header name, Standardized format, Applied Filters.
- **Power BI: (Power Query):** Removed columns, capitalized each word, re-ordered columns, changed type, used DAX for Calculated Column and Measures, used Visuals for Dashboard.

7. Calculated Columns & Measures

Calculated Measure	Description
Total Allocation	Calculates the total quantity of foodgrains allocated under the National Food Security Act (NFSA) across the selected states, months, or years. It is a key performance indicator used to monitor the total allocation of foodgrains.
Total Distribution	Calculates the total quantity of foodgrains distributed to beneficiaries through the Public Distribution System (PDS), including both ePoS and manual distribution.
Total Transactions	Calculates the total number of ration card transactions recorded during the selected period. It reflects the overall activity and utilization of the PDS.
Aadhaar Transactions	Calculates the total number of Aadhaar-authenticated ration card transactions. This measure helps evaluate the adoption of Aadhaar-based digital authentication within the distribution system.
Distribution Efficiency (%)	Measures the percentage of allocated foodgrains that were successfully distributed to beneficiaries. It is calculated as **(Total Distribution ÷ Total Allocation) × 100** and indicates the effectiveness of foodgrain distribution.
Aadhaar Authentication Rate (%)	Measures the percentage of total ration card transactions that were successfully authenticated using Aadhaar. It is calculated as **(Aadhaar Transactions ÷ Total Transactions) × 100** and reflects the level of digital authentication adoption.
ePoS Usage (%)	Calculates the percentage of foodgrains distributed through Electronic Point of Sale (ePoS) devices compared to the total foodgrain distribution. It indicates the extent of digital distribution across the Public Distribution System.
Manual Distribution (%)	Calculates the percentage of foodgrains distributed manually without using ePoS devices. This measure helps identify areas where digital infrastructure or connectivity may be limited.
Average Allocation per State	Calculates the average quantity of foodgrains allocated to each state during the selected reporting period. It enables comparison of allocation patterns across states.
Average Transactions	Calculates the average number of ration card transactions recorded during the selected period. It provides an overview of average transaction activity and beneficiary engagement within the PDS.

Calculated Column	Tool Used	Description
Distribution Gap	Power Query	Calculates the difference between the allocated and distributed quantity of foodgrains for each record. It helps identify shortages or undistributed foodgrains and supports gap analysis. **Formula:** *Food grains allocated − Distribution of food grains*.
Distribution Status	Power Query	Classifies the distribution performance based on the distribution gap. For example, records may be categorized as **Pending Distribution & Completed**, making it easier to analyse operational performance.
Aadhaar Performance	Power Query	Categorizes the Aadhaar authentication percentage into performance levels such as **Excellent, Good and Need Improvement**. This helps assess the adoption of Aadhaar-based authentication across states and reporting periods.
Year-Month	DAX	Combines the **Year** and **Month** fields into a single chronological value (e.g., **Jan 2025**, **Feb 2025**). This column is used for creating monthly trend charts and ensuring proper time-based analysis.
Quarter	DAX	Derives the calendar quarter (Q1, Q2, Q3, or Q4) from the month. It enables quarterly reporting and comparison of foodgrain allocation, transactions, and distribution performance.
Distribution Mode	DAX	Identifies the primary mode of foodgrain distribution (ePoS or Manual) based on the distribution values. This column is useful for analysing digital adoption and comparing electronic versus manual distribution methods in visualizations such as donut charts and bar charts.

8. Dashboard Visual Descriptions (Power BI)

a) Slicer – Year Filter

Purpose: The Year slicer allows users to filter the dashboard by a specific reporting year. All visuals update automatically based on the selected year.
Selected Year: All (2017–2024)
Insight: The dashboard currently displays consolidated data for all available years.

b) Line Chart – Total Distribution by Year-Month

Purpose: The Line Chart illustrates the monthly trend in foodgrain distribution across the selected time period, helping identify increases, decreases, and seasonal patterns.
Insight: Distribution remained relatively stable at approximately 8 Million MT during most of 2023. A decline is observed in early 2024, indicating lower foodgrain distribution during the selected months.

c) Clustered Bar Chart – Distribution of Foodgrains by State

Purpose: The **Clustered Bar Chart** ranks states according to the quantity of foodgrains distributed.

Current Top States (Approx.):

State	Distribution
Uttar Pradesh	**≈80 Million MT**
Bihar	**≈40 Million MT**
Maharashtra	**≈32 Million MT**
West Bengal	**≈30 Million MT**
Madhya Pradesh	**≈25 Million MT**
Rajasthan	**≈22 Million MT**
Tamil Nadu	**≈18 Million MT**

Insight:
- **Uttar Pradesh** has the highest foodgrain distribution.
- **Bihar** and **Maharashtra** are the second and third highest contributors.

d) Treemap – State-wise Ration Cards Issued

Purpose: The **Treemap** displays the proportion of ration cards issued across states. Larger rectangles indicate states with more beneficiaries.

Major States Displayed:
- West Bengal
- Uttar Pradesh
- Maharashtra
- Bihar
- Karnataka
- Madhya Pradesh
- Odisha
- Gujarat
- Tamil Nadu
- Rajasthan
- Andhra Pradesh

Insight:
- **West Bengal** and **Uttar Pradesh** occupy the largest areas, indicating a high number of ration card holders.
- States such as **Maharashtra**, **Bihar**, and **Tamil Nadu** also have substantial beneficiary populations.

e) Donut Chart – ePoS vs Manual Foodgrain Distribution

Purpose: The **Donut Chart** compares the quantity of foodgrains distributed through **Electronic Point of Sale (ePoS)** devices versus **manual distribution**.

Current Values:

Distribution Mode	Quantity	Percentage
ePoS Distribution	324 Million MT	88.68%
Manual Distribution	41 Million MT	11.32%

Insight:

- Nearly **89%** of foodgrain distribution is carried out through **ePoS**, indicating strong adoption of digital distribution systems.
- Only **11%** of distribution is manual, suggesting improved transparency and digital governance.

f) <u>Gauge Chart – Distribution Efficiency</u>

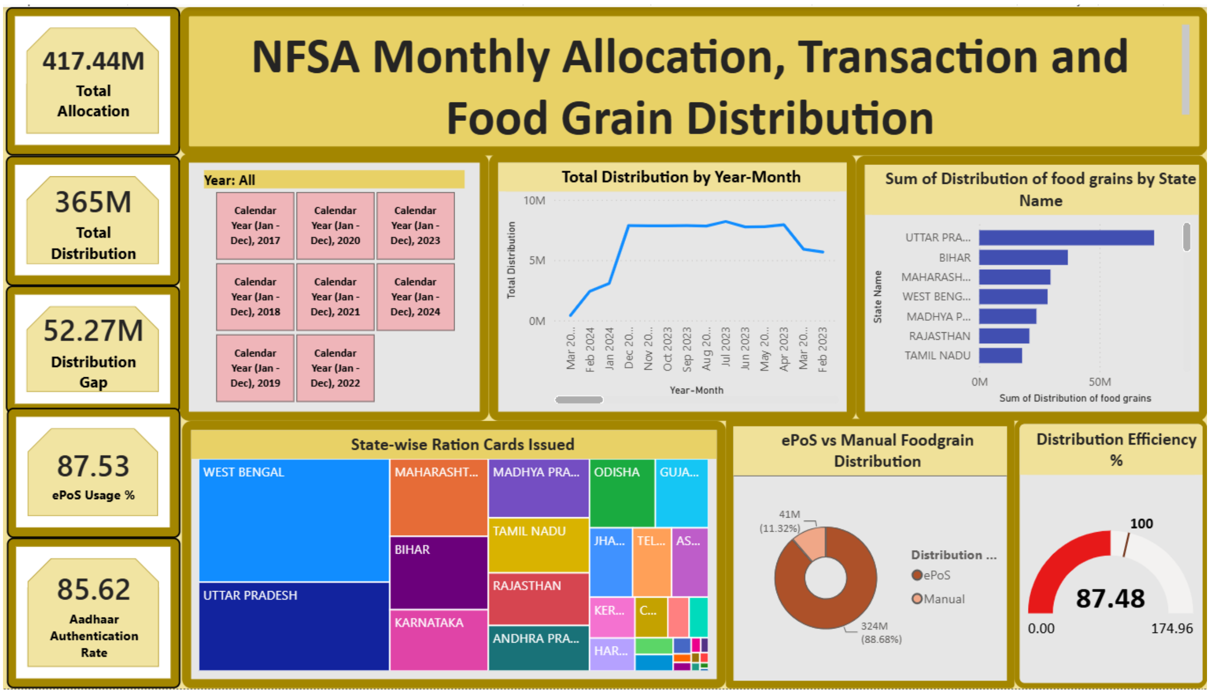
Purpose: The **Gauge Chart** measures the percentage of allocated foodgrains that were successfully distributed.

Current Values:

Metric	Value
Distribution Efficiency	87.48%
Target	100%

Insight:

- Approximately **87.48%** of allocated foodgrains have been distributed.
- The remaining **12.52%** represents the distribution gap, highlighting an opportunity to improve delivery efficiency.



9. Conclusion:

- Developed an interactive **Power BI dashboard** to analyse NFSA monthly allocation, transactions, and foodgrain distribution.
- Monitored **state-wise and monthly distribution** trends using interactive visualizations.
- Evaluated the **efficiency of foodgrain distribution** by comparing allocation with actual distribution.
- Analysed the adoption of **Aadhaar authentication** and **ePoS-based distribution** across states.
- Identified **top-performing states** based on foodgrain distribution and ration card issuance.
- Used **Power Query** and **DAX** to clean, transform, and model the data effectively.
- Implemented **KPIs, charts, maps, and slicers** to provide meaningful and interactive insights.
- Enabled **data-driven decision-making** by presenting key performance metrics in a user-friendly dashboard.
- Demonstrated how Power BI can improve **transparency, monitoring, and reporting** of the Public Distribution System (PDS).
- The project highlights the value of **Business Intelligence** in supporting efficient implementation of the National Food Security Act (NFSA).